

each user. At its peak in the mid 1990s, it offered more than 20,000 services, ranging from home banking to specialized research databases. The Minitel was in a large proportion of French homes 10 years before most Americans had ever heard of the Internet.

1.7.4 The Internet Explosion: The 1990s

The 1990s were ushered in with a number of events that symbolized the continued evolution and the soon-to-arrive commercialization of the Internet. ARPAnet, the progenitor of the Internet, ceased to exist. In 1991, NSFNET lifted its restrictions on the use of NSFNET for commercial purposes. NSFNET itself would be decommissioned in 1995, with Internet backbone traffic being carried by commercial Internet Service Providers.

The main event of the 1990s was to be the emergence of the World Wide Web application, which brought the Internet into the homes and businesses of millions of people worldwide. The Web served as a platform for enabling and deploying hundreds of new applications that we take for granted today, including search (e.g., Google and Bing) Internet commerce (e.g., Amazon and eBay) and social networks (e.g., Facebook).

The Web was invented at CERN by Tim Berners-Lee between 1989 and 1991 [Berners-Lee 1989], based on ideas originating in earlier work on hypertext from the 1940s by Vannevar Bush [Bush 1945] and since the 1960s by Ted Nelson [Xanadu 2012]. Berners-Lee and his associates developed initial versions of HTML, HTTP, a Web server, and a browser—the four key components of the Web. Around the end of 1993 there were about two hundred Web servers in operation, this collection of servers being just a harbinger of what was about to come. At about this time several researchers were developing Web browsers with GUI interfaces, including Marc Andreessen, who along with Jim Clark, formed Mosaic Communications, which later became Netscape Communications Corporation [Cusumano 1998; Quittner 1998]. By 1995, university students were using Netscape browsers to surf the Web on a daily basis. At about this time companies—big and small—began to operate Web servers and transact commerce over the Web. In 1996, Microsoft started to make browsers, which started the browser war between Netscape and Microsoft, which Microsoft won a few years later [Cusumano 1998].

The second half of the 1990s was a period of tremendous growth and innovation for the Internet, with major corporations and thousands of startups creating Internet products and services. By the end of the millennium the Internet was supporting hundreds of popular applications, including four killer applications:

- E-mail, including attachments and Web-accessible e-mail
- The Web, including Web browsing and Internet commerce

- Instant messaging, with contact lists
- Peer-to-peer file sharing of MP3s, pioneered by Napster

Interestingly, the first two killer applications came from the research community, whereas the last two were created by a few young entrepreneurs.

The period from 1995 to 2001 was a roller-coaster ride for the Internet in the financial markets. Before they were even profitable, hundreds of Internet startups made initial public offerings and started to be traded in a stock market. Many companies were valued in the billions of dollars without having any significant revenue streams. The Internet stocks collapsed in 2000–2001, and many startups shut down. Nevertheless, a number of companies emerged as big winners in the Internet space, including Microsoft, Cisco, Yahoo, e-Bay, Google, and Amazon.

1.7.5 The New Millennium

Innovation in computer networking continues at a rapid pace. Advances are being made on all fronts, including deployments of faster routers and higher transmission speeds in both access networks and in network backbones. But the following developments merit special attention:

- Since the beginning of the millennium, we have been seeing aggressive deployment of broadband Internet access to homes—not only cable modems and DSL but also fiber to the home, as discussed in Section 1.2. This high-speed Internet access has set the stage for a wealth of video applications, including the distribution of user-generated video (for example, YouTube), on-demand streaming of movies and television shows (e.g., Netflix), and multi-person video conference (e.g., Skype).
- The increasing ubiquity of high-speed (54 Mbps and higher) public WiFi networks and medium-speed (up to a few Mbps) Internet access via 3G and 4G cellular telephony networks is not only making it possible to remain constantly connected while on the move, but also enabling new location-specific applications. The number of wireless devices connecting to the Internet surpassed the number of wired devices in 2011. This high-speed wireless access has set the stage for the rapid emergence of hand-held computers (iPhones, Androids, iPads, and so on), which enjoy constant and untethered access to the Internet.
- Online social networks, such as Facebook and Twitter, have created massive people networks on top of the Internet. Many Internet users today “live” primarily within Facebook. Through their APIs, the online social networks create platforms for new networked applications and distributed games.